



12427 - MIRI Planets Orbiting White Dwarfs at Extreme Resolution

Cycle: 5, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

Thanks to its exquisite sensitivity, MIRI recently led to the first direct detections of post-main-sequence planets orbiting nearby white dwarfs. MIRI has been used to target nearly 100 nearby white dwarfs with imaging at 7.7-21 microns, with two large Cycle 2 Survey programs (MEOW and MEAD) executing the majority of these observations. With conventional imaging methods, MIRI's long wavelengths limit these images' resolution to > 10 AU, where giant planets are understood to be less abundant and where Gaia can rule out companions. We propose an archival program to enable robust MIRI super-resolution (resolution inside λ/D) by optimizing the data processing technique of kernel phase interferometry, which was recently demonstrated on MIRI for the first time. By refining the MIRI kernel phase method we will reach Jovian mass sensitivity on orbits down to and within ~ 5 AU around nearby white dwarfs, accessing parameter space where giant planets are known to be more abundant. We will process all archival MIRI white dwarf imaging using this technique, with a high probability of multiple close-in direct planet discoveries that can be used to refine post-main-sequence planetary evolution theory. We will also carry out a detailed MIRI kernel phase performance analysis and release tools that can be used by the community. These can be applied to all imaging scenarios that would benefit from moderate contrast at extreme resolution,

JWST Proposal 12427 (Created: Friday, March 13, 2026, 6:04:14PM Eastern Standard Time) - Overview

including searches for close-in companions to brown dwarfs and giant planets around young stars. This program will lead to new progress in post-main-sequence exoplanetary studies while enabling a wide range of future MIRI high resolution imaging.

OBSERVING DESCRIPTION